SS–A–674b
25 NOVEMBER 1952
SUPERSEDING
Fed. Spec. SS–A–674a
23 August 1944

FEDERAL SPECIFICATION

ASPHALT, PAVING, EMULSION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types and intended use.—Asphalt paving emulsion covered by this specification shall be of the following types, and for the respective applications indicated, as specified:

Type RS–1.—Rapid-setting, low-viscosity, emulsified asphalt for surface treatment and bituminous-macadam penetration.

Type RS–2.—Rapid-setting, high-viscosity, emulsified asphalt for surface treatment.

Type MS–1.—Low-consistency, medium-setting, emulsified asphalt; for coarse-aggregate mixes used in retread construction, with substantially all of the aggregate retained on a ⅛-inch sieve and practically no material passing the No. 200 (74-micron) sieve.

Type MS–2.—Medium-consistency, medium-setting, emulsified asphalt; for plant prepared coarse-aggregate mixes with substantially all of the aggregate retained on a ⅛-inch sieve and practically no material passing a No. 200 (74-micron) sieve.

Type MS–3.—High-consistency, medium-setting, heavy premixing grade of emulsified asphalt for use and storage above 40°F.; for plant mix or general maintenance patching; used with coarse aggregate, substantially all of which is retained on a ⅛-inch sieve and practically no material passing a No. 200 (74-micron) sieve.

Type MS–4.—High-consistency, medium-setting, heavy premixing grade of emulsified asphalt which does not separate as the result of freezing; for plant mix or general maintenance patching; used with coarse aggregate, substantially all of which is retained on a ⅛-inch sieve and practically no material passing a No. 200 (74-micron) sieve.

Type SS–1.—Slow-setting, low-viscosity, emulsified asphalt for fine-aggregate mixes, in which a substantial quantity of aggregate passes a ⅛-inch sieve and a portion may pass a No. 200 (74-micron) sieve.

Type SS–2.—Low-viscosity, slow-setting, and low-penetration emulsified asphalt for fine-aggregate mixes and for admixture with soil aggregates in which a portion passes a No. 200 (74-micron) sieve.

2. APPLICABLE SPECIFICATION

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

SS–R–406—Road and Paving Materials; Methods of Sampling and Testing.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, or cash,

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to the General Services Administration, Business Service Center, Region 3, Room 7110, Seventh and D Streets, S. W., Washington 25, D. C. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle. Prices of additional copies may be obtained from the GSA Regional Offices.)

2.2 Specifications and publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 All types, except MS–4. — Emulsified asphalts of all types, except MS–4, shall be homogeneous and shall show no separation of asphalt at the time of use provided that separation has not been caused by freezing after delivery, and provided further that this requirement shall not apply if the material is held more than 30 days after delivery.

3.2 Type MS–4. — Emulsified asphalt of type MS–4, winter grade, shall be homogeneous and shall show no separation of asphalt at time of use, provided that this requirement shall not apply if the material is held more than 30 days after delivery.

3.3 The various types of emulsified asphalt shall meet the requirements given in table I when tested as specified in 4.2.

4. SAMPLING AND TEST PROCEDURES

4.1 Sampling. — Samples of emulsified asphalt shall be taken in accordance with

TABLE I.—*Requirements for emulsified asphalt*

Tests	Rapid setting		Medium setting				Slow setting	
	RS–1	RS–2	MS–1	MS–2	MS–3	MS–4	SS–1	SS–2
TESTS ON EMULSION								
Viscosity, furol, 60 ml., 77°F., seconds	20–100		20–100	100–700			20–100	20–100
Viscosity, furol, 60 ml., 122°F., do		75–400						
Residue by distillation, percent	57–62	62–69	57–62	62–69	65+	65+	57–62	57–62
Settlement, 7 day, maximum, difference	3	3	3	3				
Demulsibility:								
50 ml. 0.10 N $CaCl_2$, percent			0–30	0–30				
35 ml. 0.02 N $CaCl_2$, do	60+	50+						
Freezing test, 3 cycles						homogeneous		
Sieve test, maximum, percent	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10
Miscibility with water, hours			2	2	2	2		
Modified miscibility[1], maximum, percent							4.5	4.5
Coating test			(2)	(2)	(2)	(2)		
Cement mixing test, maximum, percent							2.0	2.0
TESTS ON RESIDUE FROM DISTILLATION TEST								
Penetration, 77°F., 100 gm., 5 seconds	100–200	100–200	100–200	100–200	100–200	125–225	100–200	40–90
Soluble in CS_2, minimum, percent	97	97	97	97	97	97	97	97
Ash, maximum, percent	2	2	2	2	2	2	2	2
Ductility at 77°F., minimum, cm	40	40	40	40	40	40	40	40

[1] If the sample of emulsified asphalt fails to conform to the requirement for modified miscibility, the sample shall be tested for 7-day settlement and for miscibility. If the numerical difference between the average percentages of asphaltic residue in the 7-day settlement test is less than 3, and if the standard miscibility test shows no appreciable coagulation in 2 hours, the requirement for modified miscibility shall be waived.

[2] Shall show no appreciable separation in 3 minutes.

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method 101.11 described in Federal Specification SS–R–406. Samples shall be stored in clean, air-tight, sealed containers at a temperature of not less than 40°F. until tested. In the case of type MS–4 emulsified asphalt, the temperature requirement of 40°F. is not applicable. For types RS–1 and RS–2, the containers shall be of glass or black iron.

4.2 **Tests.**—Tests shall be made in accordance with the following methods described in Federal Specification SS–R–406:

Tests	Methods
Viscosity, Saybolt furol	222.01
Residue by distillation	222.01
Settlement	222.01
Demulsibility	222.01
Freezing	222.01
Sieve	222.01
Miscibility with water	222.01
Modified miscibility	222.01
Coating	222.01
Cement mixing	222.01
Penetration	214.01
Solubility in CS_2	210.0
Ash	226.0
Ductility	215.0

5. PREPARATION FOR DELIVERY

5.1 **Packaging.** — Unless otherwise specified, commercial packages are acceptable under this specification.

5.2 **Packing.** — The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**—Unless otherwise specified, shipping containers shall be marked with the name of the material, the type, and the quantity therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 **Availability.**—All types of emulsified asphalts shown herein may not be available from a single producer or from producers in a given area. The availability of a given type should be determined prior to invitation for bids.

6.1.1 *Ordering data.* — Purchasers should specify the type and quantity required and should exercise any desired options offered herein (see 1.1, 5.1, 5.3, and section 7).

6.2 **Transportation description.**

6.2.1 *Transportation description applicable to these items is:*

Asphalt (asphaltum), natural, by-product or petroleum, liquid, other than paint.
Carload weight 40,000 pounds.
Truckload weight 40,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, standards, and other publications, of the issues in effect on date of invitation for bids, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army.**

7.2.1 *Applicable specification and standard.*

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Federal Specification:

RR–D–729—Drums; Steel, Type 5B (for Liquid Petroleum Products).

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications and standards required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packing for overseas shipment.* — Unless otherwise specified in the contract or order, the asphalt shall be packed in steel drums conforming to Federal Specification RR–D–729.

7.2.3 *Marking.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

7.3 Navy.

7.3.1 *Applicable specifications, standard, and other publications.*

Navy Department Specification:

General Specifications for Inspection of Material.

Military Standard:

MIL–STD–129—Marking of Shipments.

Interstate Commerce Commission Publication:

Regulations for Transportation of Explosives and Other Dangerous Articles by Freight.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles by Freight may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

7.3.2 *Inspection procedures.* — For Naval purchases the general inspection procedures shall be in accordance with the General Specifications for Inspection of Material.

7.3.3 *For domestic shipment and storage or overseas shipment.* — Unless otherwise specified the subject commodity shall be prepared for delivery in 55-gallon drums conforming to the requirements of Interstate Commerce Commission Specification 17E.

7.3.4 *Marking.*—In addition to any special marking required by the contract or order, marking shall be in accordance with Military Standard MIL–STD–129.

NAVY INTEREST: S Y

7.4 Air Force.

7.4.1 *Applicable standard.*

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.4.1 *Marking of shipments.* — Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129. The nomenclature shall read as follows:

Asphalt, Petroleum, Paving, Emulsion
Type _____*
Fed. Spec. SS–A–674b

* Applicable data to be entered by the contractor.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

★ U. S. GOVERNMENT PRINTING OFFICE: 1952

Copies of this specification can be purchased for 5 cents each from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. Single copies required for bidding purposes are available without charge at the nearest GSA Regional Office.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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